

Md.Mubtasim Fuad

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[linkedin](#) [github](#) [Portfolio](#)



Education

North South University

Bachelor of Computer Science and Engineering. **GPA: 3.02/4.00**

Jun 2020 – Sept 2025

Dhaka, BD

Mirpur Cantonment Public School & College

Higher Secondary School Certificate (HSC). **GPA: 4.08/5.00**

May 2017 – May 2019

Dhaka, BD

Dhaka Residential Model College

Secondary School Certificate (SSC). **GPA: 5.00/5.00**

Jan 2014 – April-2016

Dhaka, BD

Cantonment Public School & College, Rangpur

Junior School Certificate (JSC). **GPA: 5.00/5.00**

Jan 2013 – Dec-2013

Rangpur, BD

Experience

Campus Ambassador

Jan 2024 – Dec 2024

Caretutors

Dhaka, BD

- Advocated for EdTech-driven academic support, promoting Caretutors to enhance digital learning adoption on campus.
- Facilitated onboarding and live demos, directly **assisting 200+ students** in navigating tutoring tools and resources.
- Collected and reported user feedback to tech teams, contributing to platform usability improvements..

Tutor

July 2021 – Present

Caretutors

Dhaka, BD

- Tutored **50+ students over 5+ years** through personalized home and online sessions in Math, Physics, and Computer Science.
- Incorporated digital tools (**Zoom, Google Meet, interactive whiteboards, coding platforms**) to create dynamic online learning experiences.
- Guided students through advanced topics, including **calculus, algorithms, and physics simulations**, with a focus on application.

Trainee Officer

January 2021 – July 2021

Mec Tech Corporation LTD.

Dhaka, BD

- Prepared **detailed plans, drawings** and the other documents for the manufacture and erection of steel member using **AutoCAD**
- Employed Advanced Excel functions (VLOOKUP, Pivot Tables) for in-depth data analysis and report creation, **reducing manual work by 2 hours** and increasing workflow efficiency.

Projects

Directed Research Bias Behind the Box: Evaluating Racial Fairness in Object Detection Models | *Python, PyTorch, CNNs,*

Data Augmentation

Summer 2025

- Engineered a modular Racial Bias Detection System to evaluate pretrained models (YOLOv11, ResNet-50, VGG-16, MobileNetV2) on demographically balanced datasets (FairFace/UTKFace)
- Utilized **transfer learning** and fine-tuning with the **Adam optimizer** on a **Kaggle GPU environment** to classify faces into seven racial classes, freezing early layers to preserve pretrained features.
- Concluded that MobileNetV2 was the fastest/lightest but had the largest bias gap (), demonstrating a critical trade-off between model efficiency and fairness for minority groups.
- Assessed model fairness by generating normalized confusion matrices and computing a bias gap based on the difference between maximum and minimum per-race accuracy.

Empty Parking Spot Detection with Real-time Footage | *Pandas, Scikit-learn, Matplotlib, Pillow*

Summer 2025

- Engineered a real-time parking spot detection system leveraging AI-powered machine learning and computer vision to accurately identify empty spaces from video footage, enhancing smart parking management
- Designed a flexible model architecture adaptable to various deep learning networks (e.g., CNN, ESPCN) and

integrated automated performance evaluation (PSNR, SSIM), utilizing PyTorch/TensorFlow, OpenCV, and NumPy for robust image analysis and data-driven optimization.

Tic_Tac_ToeAI with Minimax & Alpha-Beta Pruning | Python

Summer 2025

- Developed a Human vs. AI Tic-Tac-Toe game, implementing the Minimax algorithm with alpha-beta pruning to ensure optimal AI decision-making and strategic play.
- Integrated an optional heuristic evaluation for the AI, significantly enhancing decision-making speed and improving overall gameplay responsiveness.

SuperResAI | Python, Git, JupiterNotebook

Summer 2025

- Developed an image super-resolution system leveraging deep learning models to reconstruct high-resolution images from degraded low-resolution inputs, significantly enhancing visual clarity and detail.
- Implemented a robust methodology encompassing low-resolution image simulation, support for flexible model architectures (e.g., SRCNN, ESPCN), and automated performance evaluation (PSNR, SSIM), utilizing PyTorch/TensorFlow and OpenCV for comprehensive image quality enhancement.

Technical Skills

Languages & Frameworks	: Python, Java, C/C++, HTML/CSS, React-js, SQL, Pytorch, TensorFlow,Django,SpringBoot
Machine Learning & AI	: Pandas, Numpy, Matplotlib, Scikit-learn, LangChain, HuggingFace, Keras, Seaborn, SciPy, NLTK
Developer Tools	: AWS, GCP, Docker, Git, Cursor, Windsurf.ai , Replit, CI/CD, LLM Fine-Tuning, Prompt Engineering
Additional Softwares	: Adobe Lightroom, Adobe Premiere Pro, MS- Word, Advanced Excel, PowerPoint,AutoCAD

Personal Details

Father’s Name	: Md. Meherul Islam
Mother’s Name	: Monira Islam
Date of Birth	: 19 June, 1999
Religion	: Islam
National Id No	: 1523559605
Passport No	: BW0073900
Marital Status	: Single
Blood Group	: O+
Nationality	: Bangladeshi
Home District	: Dinajpur
Present Address	: House 340, Road 9, Block C, Bashundhara R/A, Dhaka.
Permanent Address	: Village: Valka Joypur, Post: Vagolpur, Thana: Nawabganj, District: Dinajpur.

References

Name	: Sk. Zana Alam
Title	: Managing Director
Company	: MEC Tech Corporation Ltd.
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